

UCL SGNL OPS / Orbit Dynamics Library

Assessed from the working tree at ~/odl20lite, x86-64 Linux, GCC 15

Verdict

Two corrections and one discovery. The 5.7 cm result in revision 3 was the best satellite in the best constellation; across 57 arcs the honest medians are **0.14 m for GPS and 0.33 m for Galileo**, which is not a precise orbit determination product. What did generalise is the *method* — 57 of 57 arcs converged untuned, from a cold start — and a finding nobody was looking for: **the estimated area-to-mass ratio separates the GPS constellation into hardware blocks the fit was never told about**. If that survives testing it points at a different business from the one assessed so far. It is not yet tested, and there is a live alternative explanation.

Correction to revision 3

Revision 3 led with 5.7 cm out-of-sample and treated it as the headline. Broadening the validation from 2 arcs to 57 showed that figure to be the best case rather than the typical one.

population	arcs	best	median	worst
GPS	31	0.0645 m	0.1403 m	0.4895 m
Galileo	26	0.1530 m	0.3294 m	0.5507 m
LEO	not obtained — blocked on credentialed data access			

Galileo's median is $2.3\times$ GPS's. Neither number should be quoted as 6 cm.

A second correction, smaller but worth recording because revision 3 asserted it: the claim that part of the residual was a method floor arising from velocity differentiation was **tested and found false**. Varying the velocity weight from zero to parity changes the answer not at all — the velocities are a Lagrange derivative of the same positions and carry no independent information. The conclusion against rebuilding a ray-tracer does not depend on it and still stands, but it rested on three arguments and one of them was wrong.

What did generalise

The accuracy did not. The method did, and the distinction matters because only the second was ever robustly evidenced.

57 of 57 arcs converged, in three iterations, from a cold start, across two constellations, on one deliberately untuned configuration, with no per-satellite tuning of any kind and zero failures. For a batch estimator handed unfamiliar spacecraft with no prior, that is a strong robustness result and it is the part of the Phase 2 claim that survived contact with a wider population.

Broadening also surfaced a defect that had been invisible: the SP3 reader skipped a hard-coded 22 header lines, which is correct for the format version carrying 32 GPS satellites and wrong for the newer one carrying about 120. Every non-GPS product had been failing to parse. Until it was fixed, this software *could only ever have validated against GPS* — exactly the class of defect that makes a narrow result look like a general one.

The discovery, and why it is not yet a claim

The estimated area-to-mass ratio sorts the GPS constellation into groups that correspond to hardware blocks, without being told that blocks exist.

group	estimated A/m
twelve satellites, agreeing to $\sim 1\%$ ($0.0002 \text{ m}^2/\text{kg}$)	$\sim 0.0206 \text{ m}^2/\text{kg}$
five satellites, clearly separated — attributed to the heavier GPS III vehicles	$0.0143\text{--}0.0171 \text{ m}^2/\text{kg}$
two Galileo outliers — the satellites left in eccentric orbits by the 2014 launch failure	$0.0226\text{--}0.0228 \text{ m}^2/\text{kg}$

Why this would matter more than the orbit accuracy. Fitting orbits to 0.14 m is not a product; analysis centres reach a few centimetres. But inferring a physical property of an object from its dynamics alone, precisely enough to resolve hardware classes, is the core problem in space situational awareness — estimating area-to-mass is how an unknown object gets characterised at all. That is a different capability from precise orbit determination, and it does not need the missing ray-traced grid files.

Why it cannot be claimed yet. The group with the distinct area-to-mass is also the group that fits *worst* — 0.26 to 0.37 m against 0.06 to 0.14 m for the tight cluster. A parameter absorbing unmodelled dynamics will also look like a distinct cluster. Until that is ruled out, “the fit blindly recovers hardware class” and “the fit mis-estimates one block” are the same data. Four of the five block attributions are also inferred from the very cluster being validated, which is circular.

Testing this is now the highest-value work available and it is commissioned: independent block attribution, a direct check of whether estimated area-to-mass correlates with residual size, and a comparison of the estimated ratio against the published mass ratio between blocks. **An artefact finding would be as valuable as a positive one**, and is wanted before anything is shown to anyone outside.

Where the asset stands

Revision 1	The ray-traced SRP engine — scarce, and missing its fuel
Revision 2	The validation — a credential making the SRP claim falsifiable
Revision 3	A parameter-estimating orbit determination capability
Revision 4	A robust estimator that may characterise objects from dynamics alone. The orbit accuracy is unremarkable; the estimator’s behaviour is not

Each revision moved because evidence arrived, not because the framing was rearranged. The current position is the least flattering and the best supported.

What still blocks a transaction

FOR THE USER not solvable in code

- **No licence or copyright statement anywhere in the first-party code**, which originates from UCL SGNL. **Still the single largest obstacle**, larger than any engineering gap.
- **CALCEPH is triple-licensed** and statically linked into every binary, with no choice recorded. Optional; the build drops it cleanly.
- **Credentialed data access.** The LEO validation is blocked on an Earthdata login. LEO is the regime where drag enters and where non-conservative modelling actually bites, so it is the most informative single addition remaining.

ENGINEERING

- **No observation models.** It fits to already-determined precise orbits, not to raw tracking data. If the characterisation finding survives testing, *this becomes the decisive gap*: the objects worth characterising are debris and unknown satellites, which have radar and optical tracks, not published precise ephemerides.
- **No covariance.** An estimate without an uncertainty is not usable for characterisation or for conjunction assessment.
- No manoeuvre modelling; single-threaded; no Python bindings; IAU1980 frames.

Valuation

- **As an orbit determination product: approximately £0**, and revision 4 strengthens rather than weakens that. A 0.14 m median against a few centimetres from free, better-governed alternatives is not a market position.
- **The estimator's robustness: real but not separately saleable.** 57 of 57 untuned convergences is good engineering evidence, not an offer.
- **The characterisation finding: the only thing here that could support a business, and it is unproven.** If it survives Phase 4 it is a capability with a defined market; if it is an artefact, the asset reverts to a well-documented, validated archive.
- **Replacement cost remains the wrong frame** at £300k–£600k.

Recommendation

Do not decide anything commercially until the characterisation finding is tested. That test is days of work, not months, and the answer changes which business this is.

1. **Test the block separation.** Independent attribution, the residual-correlation confound, and the published mass ratio. *Commissioned and in progress.*
2. **If it holds: build observation models.** That is what converts a tool fitting published ephemerides into one that characterises objects from real tracking data, and it is the gap between a demonstration and a service. Do not start it before the answer.
3. **If it is an artefact: stop.** The tree is then a validated, well-documented body of work with an honest accuracy figure and a robust estimator — publishable, useful for teaching, and a legitimate place to finish.
4. **Settle the IP position in parallel.** It gates every outcome and needs no engineering.

Basis: measured facts come from the tree at `/home/rog/odl20lite` and from runs executed on 15 September 2026, including Phase 1–3 work carried out by the `odl` maintainer session, spot-verified here. The competitive and valuation sections are judgement. Revision 4 supersedes revision 3, whose headline accuracy figure was a best case reported as typical and which repeated a velocity-floor claim since shown to be false.